

AI Ethics and Fairness

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Fairness is one of the most prominent values in the Ethics and Artificial Intelligence (AI) debate. This module provides an overview of the main account of fairness in AI and focuses on the need to pursue an ethical inquiry into the concept of fairness aimed at revising and enriching it, so as to design, develop and deploy really fair AI systems.

Lecture Transcript

0:02 Hello everyone, my name is Benedetta Giovanola. I hold the Jean Monnet Chair ethics for inclusive digital Europe co-funded by the European Union. I'm also a full professor of ethics at the University of Macerata in Italy and the regular visiting professors at Tufts University in the US. My main teaching and research interests are in AI ethics and digital ethics, as well as in media ethics with particular focus on digital media ethics, theories of justice and business ethics. I'm very glad to be able to join this extremely interesting initiative that is the AI Ethics: Global Perspectives course. And today, I will bring in the course the perspective on AI ethics and fairness that we are developing in the framework of the Jean Monnet chair ethics for inclusive digital Europe. So I will first provide a sort of background information regarding the aims of the chair. And then we'll focus on the specific topic of this lecture, that is AI ethics and fairness.

1:27 Now, the chair is mainly devoted to further highlight and strengthen the key role of ethics to raise awareness about the importance of values in the field of digital technologies, and especially AI technologies, AI-based technologies and systems. And to show that these ethical values are fundamental, in order to foster a human centered and ethically oriented approach to digital technology and AI that connects with, but also goes beyond, legal compliance. And that can also strengthen participation in democratic life, civic engagement, also foster equity and social inclusion in Europe and beyond. So in this broad framework, we are focusing on what we take to be the most important ethical values that could be at the core of human centered and ethically oriented AI and AI systems. And among the different values we're focusing on for today's lecture, I decided to focus on fairness and I will explain in a while why I decided to focus on it.

2:56 So this is the structure and the overview of today's course and module. First of all, I will share with you some background on fairness in AI, and in particular, highlight the main conceptions of fairness that come up in AI ethics. Then I will suggest that we need to go beyond these standard definitions, and really develop a deep idea of fairness and ethical value. And then in the last part of this lecture, I will try to discuss what it means to apply these revised notions of fairness in the design, the development and deployment of AI systems.

4:04 Now, why fairness in AI? Well, fairness is actually one of the most prominent values in AI. It is gaining an increasing attention, both in academia and so in research that is carried out by scholars in the academia, but also

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it's gaining growing attention on the side of the industry. And of course, there are reasons for that. And one of the main reasons is that more and more decisions are nowadays delegated to AI systems. Most of these systems rely on machine learning techniques. They might lead to autonomous or semi autonomous algorithms such as for example, the machine learning based algorithms, and so, a growing body of literature has been developed to focus on the need to evaluate whether these systems are contributing or not to fairness and why is it so. Because if on the one side more and more decisions are either partially or fully delegated to AI systems, on the other side, also a growing number of flows involving discrimination in AI systems have been uncovered. We know that the exponential progress and use of machine learning and more broadly speaking AI algorithms, AI-based algorithms, are very complex. They have been sort of accused to be characterized by a high degree of opacity, some claimed that they are sort of black boxes. And so, growing attention is paid not only to the issue of transparency or explicability of these systems, but also to the effect of the systems for the common good, for the social good, with a particular emphasis on fairness.

6:26 At the same time, even though fairness in AI is acknowledged as an urgent task both in academia and in industry exactly to prevent or mitigate at least risks of discrimination, there is not so much agreement on what fairness in AI is. So there are actually quite, several definitions of fairness in AI. One of the most widespread definitions is that according to which fairness is anti-classification. So it would be achievable by avoiding the use of proxies in AI systems that refer to protected categories such as, for example, the most, you know, well known ones, like race or gender, or even religion, and so on and so forth. But there are also other approaches that understand fairness as classification parity, which would mean that a system, an AI system, would be fair if the measures of its predictive performance are equal across protected groups. Also, we have come up with definitions that understand fairness as calibration. That is, fairness given by the measure of how well calibrated an algorithm is between protected groups. Or last but not least, fairness as statistical disparity, according to which the fairness of an AI based system would correspond to an equal average probability estimate over all members of protected groups. So these definitions, as you might see are quite different. And sometimes they might also be mutually incompatible. So for example, in some cases, the removal of proxies for protected categories could be beneficial. But in other cases, the consideration of protected characteristics that would be crucial, exactly to make fair decision. And in fact, in the literature, it has also been claimed that so called fair AI systems, so AI systems that would comply with the standard definitions, like the one you can see in this slide would ultimately be deeply unfair.

9:07 So, discrimination is also a key feature that comes in here. So despite the different definitions of fairness in AI, there are also some common features. So on the one side, we are, let's say in this paradoxical situation, in which the different definitions might be in contrast or even in conflict with one another, but on the other side, there are also some common features. And I would say that the most common features of the mainstream understandings of fairness in AI is the focus on the consideration or treatment of protected groups or categories that we have seen spans across all the four definitions we analyzed before just as exemplary cases of standard definitions. And another key feature is a strong focus on discrimination. So basically, fairness in this discussion is understood mostly as non-discrimination. And non-discrimination, in turn, is understood mostly as non-discrimination against protected groups or categories that are indeed traced back to the most common understanding of what amounts to be protected categories. So, for example, race, gender, and so on and so forth. So, as a consequence, most of the attention has been paid to even technical ways aimed at removing this discrimination against protected groups or categories. And the main focus in these attempts has been the focus to

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remove bias. So the underlying idea would be that guaranteeing, ensuring non-discrimination would require eliminating, or at least mitigating, bias. And then in fact, in the debate on what is usually labeled as algorithmic fairness, huge attention is exactly paid to the need of non non-discriminating through the removal of persistent and crucial biases.

11:48 So to sum up on the standard definitions of fairness in AI. I think despite differences, what comes up is that fairness in AI is mostly understood as negative fairness. I'm here clearly rephrasing the sort of standard, or at least very well known, our discussion of fairness in AI. And what comes out from the analysis of different and sometimes mutually incompatible definitions of fairness, we can highlight that fairness in AI is mostly understood as negative fairness. Here in labeling, fairness as negative fairness, so I'm rephrasing the well known distinction that was first introduced by Isaiah Berlin, in his studies on liberty, where he distinguishes a negative freedom from a positive freedom. A negative freedom would be freedom understood as the absence of obstacles, or to put it in another way, a freedom from something. Whereas positive freedom would be understood as the freedom to do something or some that has claimed freedom for being a certain kind of person through free actions that would be performed. And I think this distinction is particularly interesting applied to our discussion on fairness, because it would allow us to highlight that fairness in the discussion on AI systems is mostly understood as the absence of something. So this is the meaning of negative fairness, fairness understood as the absence of discrimination. So furnace as a prevention from discrimination or the abolishment of discrimination, or at least the mitigation of discrimination.

14:16 At the same time, however, it seems that these definitions of fairness is not enough, because if on the one side, we can agree that discrimination is a key component of fairness. On the other side, as we also have highlighted, just the overlap between fairness and this non-discrimination might eventually lead to unfair outcomes. And this is exactly why understanding fairness just as non-discrimination is not enough. So what I'm proposing here is to try on the one side acknowledge the link, the relationship between fairness and discrimination, but on the other side, try to go beyond that. And how can we do so? We can do so by developing a stronger understanding of fairness as an ethical value. So really drawing insights from the understanding of fairness in the debate on moral philosophy, ethics and public ethics, also theories of justice.

15:36 Now, if we look at this discussion, what comes up is that on one side, fairness and discrimination are related. This is especially because fairness would require treating people as equals. It would also require not all need distributing something, let's say fairly, but it would also require to prompt relationships among persons that relate to one another and recognize one another as equals. This is quite a standard understanding of fairness in the most prominent positions in the contemporary debate on theories of justice starting from John Rawls, and then also in the following debate in the framework of especially liberal egalitarian perspectives. At the same time, treating people as equals and prompting or fostering a society of equals as some scholars have claimed, would also require not to degrade anyone, not to treat anyone with cruelty, not humiliating anyone in such a way that would undermine their capacity to develop and maintain an integral sense of the self. So these would also amount to non-discriminating anyone because discriminating someone, as for example, belonging to some groups or categories, would amount to hinder or even prevent an equal treatment of peoples and the achievement of a society of equals.

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17:39 Now, as we can see already from here, there's a strong ethical underpinning of discrimination and fairness, they both have an ethical significance. And so usually what is criticized is not, let's say, discrimination as such, but wrongful discrimination. So that kind of discrimination that would bring about moral disrespect. So the moral disrespect would in turn amount to humiliating people or degrading them or not treating them as equals. And fairness in this regard, would imply, to provide more work against wrongful discrimination, exactly due to the acknowledgement of moral equality among people.

18:30 So if this is the connection between fairness and discrimination, then we can highlight also, drawing insights from the major discussions on the topics in ethics, and in theories of justice. These kinds of studies also allow us to highlight that there's also a difference between fairness and discrimination. And the most important difference is so to say a wider scope of fairness, which entails both a distributive and a soulful relational dimension. So that would require both to establish the condition so that people would actively recognize one another as equal, which extends beyond not just discriminating them. And fairness would also require a distributive dimension. This is one of the main features in the debate on theories of justice, that would in turn, be based on this idea of the need of treating people as equals. So treating people as equals not only not to discriminate them, but we might say to empower them. And here come in some interesting ideas that are very useful for us to develop a deeper understanding of fairness as an ethical value.

20:08 First of all, one of the key principles that is acknowledged on the philosophical debate on fairness is the importance of a fair equality of opportunities. The fair equality of opportunities as it is usually understood, it regulates the distribution of benefits and burdens of social cooperation. Also the arrangement of socio economic inequalities in such a way that does not only prevent discrimination, but also create the conditions that enable personal agency and self realization. And that's what I meant before when it was claiming to empower, to foster the active dimension of people's capabilities and people's opportunities. So, this in turn is also grounded on the need to respect persons as persons. So to acknowledge the equal moral roles of persons in such a way that fair chances, fair effective opportunities, not just in the formal sense, but in the real ones, are really guaranteed to each and every person. At the same time, this moral understanding of equal respect for persons as persons, it's also important to highlight another key dimension of the contemporary debate on fairness in philosophical terms, which is the right to justification, which is to say, the fact that is framed in terms of an ethical demand, that no relations should exist, that cannot be justified in an adequate way towards those involved. So, these were two justification points to the importance on one side of interest objectives and also social relations and structures. But it also requires that every person is protected and empowered in their capacities to make up their own minds, to freely choose on issues of concern. So, this actually, it has been claimed as on the principle, on general and reciprocal justification. So on the fact that every person should be equally respected as a person that is as a subject, who offers and demands justification and this is also why the right to justification can also be seen as a question of power. So the question of who decides what, and it highlights a social relational component of fairness, that is intertwined with the importance of mutual recognition and also to the need to mitigate asymmetries of power, which is also a fundamental topic and the debate on AI.

23:23 So, if you look at these philosophical discussions on fairness in the framework of ethics and theories of justice, we highlight something that truly extends beyond discrimination, which is fair equality of opportunity, and the right to diversification. But there is also more. So the idea that if you really look deeper at this debate, there's also something else, which is extremely important, that's this - the fair equality of relationships. Now, the

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principles we discussed, so far, that is the fair equality of opportunity and the right to justifications as we highlighted, they both rely upon the principle of respect for persons as persons, as equal persons or equal respect for persons as persons. But as it has also been highlighted, there's not just this kind of, what is called recognition respect, so respect for persons as persons, there is also another very important dimension, which is the dimension of respect for persons as particular individuals. And this is the dimension that as we will see, in a while, turns out to be extremely important also for the debate on fairness, in AI and for its revision. So, what does respect for persons as particular individuals mean? Well, it means that if on the one side it is important to confer equal respect, as a sort of ethical response to each individual's standing as a person, so the equal respect. On the other side, it's not only important to respect people's moral values in terms of their, say like abstract capacity for agency, as it is as required by equal respect. People's moral value is not only attached, even though surely it is to their say abstract capacity for agency, but it is also attached to their particular status as particular individuals. It is different individuals that exercise their concrete agency in different and concrete ways.

25:58 So, the respect for persons as particular individuals or particular agents allows us to take into account the different ways in which different persons can exercise their agencies. And so, particular individuals are not just abstract entities, they are concrete persons, so to say, real persons that actually can be fully respected, if they are also respected in their capacity to find their attachments and affiliations, to define their commitments and ground products. And this is a different dimension from the dimension of the abstract self. Now this dimension of the particular individual, and so the respect for the particular individual, is not less important than the dimension of respect for persons as persons. And why is it so? Well it is so, because, people, so particular individuals, develop their attachments and their affiliations and their ideas of the good depending on the specific person they are. And so even the commitments individuals have or their affiliations, they involve robust intentions that can be central to one's life plans and sense of the self. And many of these commitments, including our relationships and including attachments and affiliations, are grounded, let's say, on what really matters for us. And in turn, they trigger shared intentions, capabilities for joint actions. All features that are really fundamental to understand and to achieve fairness, if we want to understand fairness, not only as something abstract, but as something concrete and attainable.

28:18 And many of these relationships are more and more shaped by AI systems and technologies. If we think for example, of all the well known phenomena of the filter bubbles or the echo chambers that aggregate news, select relevant information, also prompt relationships. These are highly shaped by the social media, which in turn are based on AI technologies. Sometimes people end up making choices without even really realizing the underlying reasons for their choices, because they are increasingly nudged and prompted by the AI systems in such ways that as it has been uncovered, might not really foster what here I call a fair equality or relationship but rather polarization attitudes, which are more and more insular. It could even lead to strengthen motivationally bias beliefs that, in turn, would lead to the strengthening of self enclosed groups that might conflict with other groups. So if we don't want that to happen, because this would be really detrimental for fairness, we have to make sure that the fair equality of relationship is also guaranteed, which would amount to say that important, they would amount to acknowledge the importance of genuine share intentions, attachments, affiliations and joint commitments. And would also acknowledge how all these are important to foster interpersonal and social relationships that can effectively contribute to the advancement of fairness.

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30:41 So to sum up, if we acknowledge the fair equality of opportunity, the right to justification, and the fair equality of relationship, we can develop an understanding of fairness as positive fairness. So fairness that would not only require and, of course, will not overlap with non-discrimination, but would really empower people and also prompt them to have genuine relationships that would contribute also to the social good and to fairer and more inclusive society. So what does this all mean for the discussion of fairness in AI? So how can this idea of fairness as an ethical value that is as positive fairness be interesting and useful for not only the discussion on fairness and AI, but also for the design, development and deployment of effectively fair AI systems? Well, first of all, focusing on the principles of fair equality of opportunities would allow to highlight the importance of not just focusing on bias, and on the removal of bias. It seems fair equality of opportunities, as we have highlighted, implies empowering people, providing them real chances, then, ensuring fairness in AI systems would also require AI systems to really provide real chances for every person. This might lead to different strategies. Of course, it's very important if we endorse, for example, an ethics by design approach. For example, designers could introduce compensatory correlations in the supervised learning phase of the AI systems to counteract or at least mitigate social and economic inequalities that are deeply rooted in our societies.

33:08 Then the right to justification. Well, the right to justification, as we have highlighted, acknowledges and highlights the status of an equal end-setter for each and every person. Now, how is it possible to set our ends? How is it possible to really freely make up our own minds and make decisions that are effectively and not just formally free? Well, if we are at least able to interpret and to understand what is going on in AI systems, for example, how we are treated, how we are nudged, how we are prompted, this does not necessarily amount to providing a full, let's say transparency of the systems, but ties in with widely debated topic of explainable AI systems. So there's actually I mean, if we focus on the right to justification, we can find really strong grounds to claim for each and every person's right to interpretable and understandable explanations of how they are treated by the AI system to really understand whether they are treated in such a way that interferes with their right to justification and capabilities to make up their own minds or not.

34:32 And last but not least, fair equality of relationships. Now focusing on the principles of the fair of equality of relationships allows us to focus on a farther social relational dimension of fairness. So it would also imply and allow us to go beyond notes and correlations, because of course, the way in which we are profiled and categorized is always depending on our belonging to specific groups that are established and fostered by the AI systems. But these in turn, as anticipated before, might also lead to a restriction of our relationships rather than a broadening or rather than allowing us to really go for genuine relationships. In some cases, it might also, you know, hinder our autonomy in choosing in ways of course, we're not fully aware of, and choosing what kind of relationships we might, we might have. So guaranteeing the principles of fair equality of relationship in AI systems would require the subjects to have information about profilation and categorization. Of course, this is not meant to be exhaustive. But these are some directions we might go for, in order to really design, develop and deploy, effectively fair AI systems.

36:30 So to sum up, fairness in AI revised as an ethical value can become a weapon of moral construction. Now, of course, here I'm paraphrasing the very influential book by Cathy O'Neill, titled "Weapons of Math Destruction", highlighting many biases, and many ethically crucial issues that are deeply embedded in AI-based technologies, and especially in algorithms. And so here the idea is that if we take fairness seriously, if we have a quite comprehensive or less complex understanding of fairness and ethical values, we might turn the AI systems

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in weapons of moral construction rather than “weapons of math destruction”. This, of course highlights the importance of an ethics by design approach. This is also something we're trying to develop in the Jean Monnet chair ethics for inclusive digital Europe. But then of course, there are also other dimensions that are important to take into account last but not least, for example, the dimension of the governance of the AI.

37:48 So I hope I provided at least some food for thought in this extremely hot topic of fairness in AI. So the lecture was meant to both provide some theoretical insights and some tracks for potential effective deployment and application of fairness as an ethical value in AI systems in such a way that AI systems can really become a force for good and can really contribute to greater fairness and to a better society. Thank you very much for your attention.